

```
In [24]: import duckdb
import pandas as pd
```

```
In [25]: con = duckdb.connect(r"C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analys
print(con.execute("SHOW TABLES").df())
```

```
          name
0  mart_ab_organizations
1      mart_ab_ridings
2  mart_bc_organizations
3      mart_bc_ridings
4  mart_on_organizations
5      mart_on_ridings
6          raw_csjs
7          stg_csjs
```

```
In [26]: on_ridings = con.execute("SELECT * FROM mart_on_ridings").df()
on_orgs   = con.execute("SELECT * FROM mart_on_organizations").df()
bc_ridings = con.execute("SELECT * FROM mart_bc_ridings").df()
bc_orgs   = con.execute("SELECT * FROM mart_bc_organizations").df()
ab_ridings = con.execute("SELECT * FROM mart_ab_ridings").df()
ab_orgs   = con.execute("SELECT * FROM mart_ab_organizations").df()
```

```
In [27]: print("ON ridings:", on_ridings.shape)
print("ON orgs:", on_orgs.shape)
print("BC ridings:", bc_ridings.shape)
print("BC orgs:", bc_orgs.shape)
print("AB ridings:", ab_ridings.shape)
print("AB orgs:", ab_orgs.shape)
```

```
ON ridings: (121, 29)
ON orgs: (3630, 29)
BC ridings: (42, 29)
BC orgs: (1260, 29)
AB ridings: (35, 29)
AB orgs: (1021, 29)
```

```
In [28]: on_ridings.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 121 entries, 0 to 120
Data columns (total 29 columns):
 #   Column                                Non-Null Count  Dtype
---  -
 0   constituency                         121 non-null    object
 1   total_number_of_organizations        121 non-null    int64
 2   funding_2017                        121 non-null    float64
 3   jobs_2017                           121 non-null    float64
 4   avg_salary_2017                     121 non-null    float64
 5   funding_2018                        121 non-null    float64
 6   jobs_2018                           121 non-null    float64
 7   avg_salary_2018                     121 non-null    float64
 8   funding_2019                        121 non-null    float64
 9   jobs_2019                           121 non-null    float64
10   avg_salary_2019                     121 non-null    float64
11   funding_2020                        121 non-null    float64
12   jobs_2020                           121 non-null    float64
13   avg_salary_2020                     121 non-null    float64
14   funding_2021                        121 non-null    float64
15   jobs_2021                           121 non-null    float64
16   avg_salary_2021                     121 non-null    float64
17   funding_2022                        121 non-null    float64
18   jobs_2022                           121 non-null    float64
19   avg_salary_2022                     121 non-null    float64
20   funding_2023                        121 non-null    float64
21   jobs_2023                           121 non-null    float64
22   avg_salary_2023                     121 non-null    float64
23   funding_2024                        121 non-null    float64
24   jobs_2024                           121 non-null    float64
25   avg_salary_2024                     121 non-null    float64
26   funding_2025                        121 non-null    float64
27   jobs_2025                           121 non-null    float64
28   avg_salary_2025                     121 non-null    float64
dtypes: float64(27), int64(1), object(1)
memory usage: 27.5+ KB

```

```
In [29]: on_ridings.head(100)
```

Out[29]:

	constituency	total_number_of_organizations	funding_2017	jobs_2017	avg_sal
0	Ajax	252	578737.0	192.0	301
1	Algoma - Manitoulin - Kapuskasing	532	577524.0	281.0	205
2	Aurora - Oak Ridges - Richmond Hill	254	498342.0	182.0	273
3	Barrie - Innisfil	215	497071.0	190.0	261
4	Barrie - Springwater - Oro- Medonte	335	590532.0	194.0	304
...	...	...	...	...	...
95	Scarborough North	296	662934.0	239.0	277
96	Scarborough Southwest	250	655290.0	197.0	332
97	Simcoe - Grey	413	507049.0	169.0	300
98	Simcoe North	364	563636.0	276.0	204
99	Spadina - Fort York	534	921978.0	265.0	347

100 rows × 29 columns



In [30]: on_orgs.head(100)

Out[30]:

	constituency	org_name	funding_2017	jobs_2017	avg_salary_2017	funding_2018
0	Essex	Town of Essex	16753.0	10.0	1675.300000	
1	Essex	The Town of LaSalle	0.0	0.0	NaN	
2	Essex	Friends of Fort Malden	42562.0	12.0	3546.833333	44
3	Essex	Municipality of Lakeshore	0.0	0.0	NaN	
4	Essex	Kiwanis Club of Windsor	0.0	0.0	NaN	
...	...	...	...	...	...	...
95	Barrie - Innisfil	Connexus Church	31558.0	12.0	2629.833333	
96	Barrie - Innisfil	Chabad Jewish Centre of Barrie	17710.0	6.0	2951.666667	14
97	Barrie - Innisfil	Recreation and Culture Services	0.0	0.0	NaN	
98	Barrie - Innisfil	Innisfil Community Church	27762.0	8.0	3470.250000	
99	Barrie - Innisfil	Emmanuel Baptist Church	21546.0	5.0	4309.200000	

100 rows × 29 columns



```

In [31]: # 第一个任务, 把所有的avg_salary全部转成保留两位小数形式!
avg_cols = [col for col in on_ridings.columns if 'avg_salary' in col]
on_ridings[avg_cols] = on_ridings[avg_cols].round(2)

avg_cols = [col for col in on_orgs.columns if 'avg_salary' in col]
on_orgs[avg_cols] = on_orgs[avg_cols].round(2)

avg_cols = [col for col in bc_ridings.columns if 'avg_salary' in col]
bc_ridings[avg_cols] = bc_ridings[avg_cols].round(2)

avg_cols = [col for col in bc_orgs.columns if 'avg_salary' in col]
bc_orgs[avg_cols] = bc_orgs[avg_cols].round(2)

avg_cols = [col for col in ab_ridings.columns if 'avg_salary' in col]
ab_ridings[avg_cols] = ab_ridings[avg_cols].round(2)

```

```
avg_cols = [col for col in ab_orgs.columns if 'avg_salary' in col]
ab_orgs[avg_cols] = ab_orgs[avg_cols].round(2)
```

In [32]: `on_ridings.head()`

Out[32]:

	constituency	total_number_of_organizations	funding_2017	jobs_2017	avg_sala
--	--------------	-------------------------------	--------------	-----------	----------

0	Ajax	252	578737.0	192.0	
---	------	-----	----------	-------	--

1	Algoma - Manitoulin - Kapusking	532	577524.0	281.0	
---	---------------------------------------	-----	----------	-------	--

2	Aurora - Oak Ridges - Richmond Hill	254	498342.0	182.0	
---	--	-----	----------	-------	--

3	Barrie - Innisfil	215	497071.0	190.0	
---	----------------------	-----	----------	-------	--

4	Barrie - Springwater - Oro- Medonte	335	590532.0	194.0	
---	--	-----	----------	-------	--

5 rows × 29 columns



In [33]: `stg_csj = con.execute("SELECT * FROM stg_csj").df()
stg_csj.head()`

Out[33]:

	year	region	constituency	org_name	amount	jobs
--	------	--------	--------------	----------	--------	------

0	2025	Alberta	Banff - Airdrie	Airdrie & District Soccer Association	31621	7
---	------	---------	--------------------	--	-------	---

1	2025	Alberta	Banff - Airdrie	Airdrie Chamber of Commerce	9341	2
---	------	---------	--------------------	-----------------------------	------	---

2	2025	Alberta	Banff - Airdrie	Alberta Institute for Wildlife Conservation	74421	16
---	------	---------	--------------------	--	-------	----

3	2025	Alberta	Banff - Airdrie	Ascend Pediatrics	1800	1
---	------	---------	--------------------	-------------------	------	---

4	2025	Alberta	Banff - Airdrie	Banff Sport Medicine Foundation	4708	1
---	------	---------	--------------------	---------------------------------	------	---

STEP 1 先把用于Tableau可视化的数据集准备好! 做好相关格式修改!

In [34]: `# 整体做清理, 把省份column加上去! 优先尝试把六大表合并成两大类大表并带省份!
on_ridings['province'] = 'Ontario'
on_orgs['province'] = 'Ontario'
bc_ridings['province'] = 'British Columbia'
bc_orgs['province'] = 'British Columbia'
ab_ridings['province'] = 'Alberta'
ab_orgs['province'] = 'Alberta'`

In [35]: `on_ridings.head(3)`

Out[35]:

	constituency	total_number_of_organizations	funding_2017	jobs_2017	avg_sala
0	Ajax	252	578737.0	192.0	
1	Algoma - Manitoulin - Kapuskasing	532	577524.0	281.0	
2	Aurora - Oak Ridges - Richmond Hill	254	498342.0	182.0	

3 rows × 30 columns



In [36]: `# 所有ridings/orgs表格拼接到一起, 是表示所有省份下所有信息!`
`all_ridings = pd.concat([on_ridings, bc_ridings, ab_ridings], ignore_index=True)`
`all_orgs = pd.concat([on_orgs, bc_orgs, ab_orgs], ignore_index=True)`

In [37]: `print("all_ridings:", all_ridings.shape)`
`print("all_orgs:", all_orgs.shape)`

all_ridings: (198, 30)
 all_orgs: (5911, 30)

In [38]: `all_ridings_long = pd.melt(all_ridings,`
`id_vars = ['constituency', 'province', 'total_number_`
`var_name = 'metric_year',`
`value_name = 'value')`

In [39]: `print(all_ridings_long.head(10))`
`print(all_ridings_long.shape)`

```

constituency province \
0 Ajax Ontario
1 Algoma - Manitoulin - Kapuskasing Ontario
2 Aurora - Oak Ridges - Richmond Hill Ontario
3 Barrie - Innisfil Ontario
4 Barrie - Springwater - Oro-Medonte Ontario
5 Bay of Quinte Ontario
6 Beaches - East York Ontario
7 Brampton Centre Ontario
8 Brampton East Ontario
9 Brampton North Ontario

total_number_of_organizations metric_year value
0 252 funding_2017 578737.0
1 532 funding_2017 577524.0
2 254 funding_2017 498342.0
3 215 funding_2017 497071.0
4 335 funding_2017 590532.0
5 577 funding_2017 730243.0
6 298 funding_2017 548435.0
7 249 funding_2017 765708.0
8 164 funding_2017 420908.0
9 137 funding_2017 492432.0
(5346, 5)

```

```

In [40]: # 将metric列分开!
all_ridings_long[['metric', 'year']] = all_ridings_long['metric_year'].str.rsplit

# 删掉原来的metric_year!
# all_ridings_long = all_ridings_long.drop(columns='metric_year')

# 再pivot, 让fundings/job/salary重新分为三列!
all_ridings_long = all_ridings_long.pivot_table(
    index = ['constituency', 'province', 'total_number_of_organizations', 'year']
    columns = 'metric',
    values = 'value'
).reset_index()

print(all_ridings_long.head(5))
print(all_ridings_long.shape)

```

```

metric constituency province total_number_of_organizations year \
0 Abbotsford British Columbia 208 2017
1 Abbotsford British Columbia 208 2018
2 Abbotsford British Columbia 208 2019
3 Abbotsford British Columbia 208 2020
4 Abbotsford British Columbia 208 2021

metric avg_salary funding jobs
0 3653.07 537001.0 147.0
1 2690.23 382013.0 142.0
2 4243.39 572858.0 135.0
3 4154.58 531786.0 128.0
4 3773.36 792406.0 210.0
(1782, 7)

```

```
In [41]: all_ridings_long
```

Out[41]:

metric	constituency	province	total_number_of_organizations	year	avg_salary	
0	Abbotsford	British Columbia	208	2017	3653.07	
1	Abbotsford	British Columbia	208	2018	2690.23	
2	Abbotsford	British Columbia	208	2019	4243.39	
3	Abbotsford	British Columbia	208	2020	4154.58	
4	Abbotsford	British Columbia	208	2021	3773.36	
...	...	...	...	...	...	
1777	York South - Weston	Ontario	243	2021	4503.46	1
1778	York South - Weston	Ontario	243	2022	4554.02	1
1779	York South - Weston	Ontario	243	2023	4474.43	
1780	York South - Weston	Ontario	243	2024	4866.81	
1781	York South - Weston	Ontario	243	2025	4922.34	1

1782 rows × 7 columns



In [43]:

```
# 现在来做org表格!  
all_orgs
```

Out[43]:

	constituency	org_name	funding_2017	jobs_2017	avg_salary_2017	fundi
0	Essex	Town of Essex	16753.0	10.0	1675.30	
1	Essex	The Town of LaSalle	0.0	0.0	NaN	
2	Essex	Friends of Fort Malden	42562.0	12.0	3546.83	
3	Essex	Municipality of Lakeshore	0.0	0.0	NaN	
4	Essex	Kiwanis Club of Windsor	0.0	0.0	NaN	
...	...	...	...	...	...	...
5906	Sherwood Park - Fort Saskatchewan	Trinity Baptist Church	9463.0	2.0	4731.50	
5907	Sherwood Park - Fort Saskatchewan	Town of Whitecourt	0.0	0.0	NaN	
5908	Sherwood Park - Fort Saskatchewan	The Hastings Lake Lutheran Bible Camp Association	0.0	0.0	NaN	
5909	Sherwood Park - Fort Saskatchewan	Shiloh	0.0	0.0	NaN	
5910	Sherwood Park - Fort Saskatchewan	Seasonal Impact Contracting Ltd.	4392.0	3.0	1464.00	

5911 rows × 30 columns



```

In [44]: all_orgs_long = pd.melt(
    all_orgs,
    id_vars=['constituency', 'province', 'org_name'],
    var_name='metric_year',
    value_name='value'
)

all_orgs_long[['metric', 'year']] = all_orgs_long['metric_year'].str.rsplit('_', 1)

all_orgs_long = all_orgs_long.pivot_table(
    index=['constituency', 'province', 'org_name', 'year'],
    columns='metric',
    values='value'
)

```

```
).reset_index()

print(all_orgs_long.head(5))
print(all_orgs_long.shape)
```

	metric	constituency	province	org_name	year	avg_salary	\
0		Abbotsford	British Columbia	ABFM Thrift Store	2017	NaN	
1		Abbotsford	British Columbia	ABFM Thrift Store	2018	NaN	
2		Abbotsford	British Columbia	ABFM Thrift Store	2019	NaN	
3		Abbotsford	British Columbia	ABFM Thrift Store	2020	NaN	
4		Abbotsford	British Columbia	ABFM Thrift Store	2021	NaN	

	metric	funding	jobs
0		0.0	0.0
1		0.0	0.0
2		0.0	0.0
3		0.0	0.0
4		0.0	0.0

(53199, 7)

In [45]: all_orgs_long

Out[45]:

	metric	constituency	province	org_name	year	avg_salary	funding	jobs
0		Abbotsford	British Columbia	ABFM Thrift Store	2017	NaN	0.0	0.0
1		Abbotsford	British Columbia	ABFM Thrift Store	2018	NaN	0.0	0.0
2		Abbotsford	British Columbia	ABFM Thrift Store	2019	NaN	0.0	0.0
3		Abbotsford	British Columbia	ABFM Thrift Store	2020	NaN	0.0	0.0
4		Abbotsford	British Columbia	ABFM Thrift Store	2021	NaN	0.0	0.0
...		...	...	...	...	...	...	...
53194		York South - Weston	Ontario	Weston Village Senior Centre	2021	3990.0	11970.0	3.0
53195		York South - Weston	Ontario	Weston Village Senior Centre	2022	4200.0	12600.0	3.0
53196		York South - Weston	Ontario	Weston Village Senior Centre	2023	4340.0	13020.0	3.0
53197		York South - Weston	Ontario	Weston Village Senior Centre	2024	NaN	0.0	0.0
53198		York South - Weston	Ontario	Weston Village Senior Centre	2025	4816.0	19264.0	4.0

53199 rows × 7 columns

```
In [46]: # 导出三个CSV来tableau做图可视化分析!
# 1. Tableau用的riding长表
all_ridings_long.to_csv(
    r'C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analysis\tableau_riding'
    index=False
)

# 2. Tableau用的org长表 (TOP30)
all_orgs_long.to_csv(
    r'C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analysis\tableau_orgs_t'
    index=False
)

# 3. 全量数据 (stg_csjs, 所有组织)
stg_csjs.to_csv(
    r'C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analysis\tableau_all_or'
    index=False
)
```

```
)

print("三个CSV导出完成！")
print("1. tableau_ridings_long.csv")
print("2. tableau_orgs_top30_long.csv")
print("3. tableau_all_orgs_full.csv")
```

三个CSV导出完成！

1. tableau_ridings_long.csv
2. tableau_orgs_top30_long.csv
3. tableau_all_orgs_full.csv

到此步骤，专门用于TABLEAU画图三个CSV导出完成！全部做好！###

现在开始正式整SQL不擅长的地方，专门用PYTHON对SQL导出的数据集做精细化清洗和格式修正！属于python能做的真正价值！

2. 导出专业Excel (6个sheet) → 自动命名sheet → 表头加粗 → 冻结首行 → 列宽自适应
3. 给amount列加货币格式 → 578737 → \$578,737
4. 给每张表加一个provincial summary行 → 每个省的选区汇总底部加一行Total
5. 简单趋势预测 → 用线性回归预测2026年funding → 加到Excel里作为额外sheet

In [47]: `pip install openpyxl`

Requirement already satisfied: openpyxl in e:\pyhcarm\lib\site-packages (3.1.5)
 Requirement already satisfied: et-xmlfile in e:\pyhcarm\lib\site-packages (from openpyxl) (1.1.0)
 Note: you may need to restart the kernel to use updated packages.

In [48]: `pip install openpyxl scikit-learn`

Requirement already satisfied: openpyxl in e:\pyhcarm\lib\site-packages (3.1.5)
 Requirement already satisfied: scikit-learn in e:\pyhcarm\lib\site-packages (1.6.1)
 Requirement already satisfied: et-xmlfile in e:\pyhcarm\lib\site-packages (from openpyxl) (1.1.0)
 Requirement already satisfied: numpy>=1.19.5 in e:\pyhcarm\lib\site-packages (from scikit-learn) (2.4.2)
 Requirement already satisfied: scipy>=1.6.0 in e:\pyhcarm\lib\site-packages (from scikit-learn) (1.17.0)
 Requirement already satisfied: joblib>=1.2.0 in e:\pyhcarm\lib\site-packages (from scikit-learn) (1.4.2)
 Requirement already satisfied: threadpoolctl>=3.1.0 in e:\pyhcarm\lib\site-packages (from scikit-learn) (3.5.0)
 Note: you may need to restart the kernel to use updated packages.

In [53]: `from openpyxl import load_workbook`
`from openpyxl.styles import Font, PatternFill, Alignment`
`from openpyxl.utils import get_column_letter`

`output_path = r'C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analysis\CSJ_`
`# 定义6个sheet的配置`

```

sheets = {
    'ON ridings': on_ridings,
    'ON organizations': on_orgs,
    'BC ridings': bc_ridings,
    'BC organizations': bc_orgs,
    'AB ridings': ab_ridings,
    'AB organizations': ab_orgs
}

# Step 1: 用pandas先导出基础数据
with pd.ExcelWriter(output_path, engine='openpyxl') as writer:
    for sheet_name, df in sheets.items():
        df.to_excel(writer, sheet_name=sheet_name, index=False)

# Step 2: 用openpyxl加格式
wb = load_workbook(output_path)

for sheet_name in wb.sheetnames:
    ws = wb[sheet_name]

    # 表头加粗 + 背景色 + 字体颜色
    header_font = Font(bold=True, color='FFFFFF', name='Arial')
    header_fill = PatternFill('solid', start_color='2F5597') # 深蓝色

    for cell in ws[1]:
        cell.font = header_font
        cell.fill = header_fill
        cell.alignment = Alignment(horizontal='center', wrap_text=True)

    # 冻结首行
    ws.freeze_panes = 'A2'

    # 列宽自适应
    for col in ws.columns:
        max_length = 0
        col_letter = get_column_letter(col[0].column)
        for cell in col:
            if cell.value:
                max_length = max(max_length, len(str(cell.value)))
        ws.column_dimensions[col_letter].width = min(max_length + 2, 30)

    # 货币格式: 所有funding列
    for row in ws.iter_rows(min_row=2):
        for cell in row:
            col_name = ws.cell(1, cell.column).value
            if col_name and 'funding' in str(col_name):
                cell.number_format = '$#,##0'
            elif col_name and 'avg_salary' in str(col_name):
                cell.number_format = '$#,##0.00'

wb.save(output_path)
print("专业Excel导出完成! ")

...

**代码做了什么:**
- 深蓝色表头 + 白色加粗字体
- 冻结首行, 滚动时表头不动
- 列宽自适应, 最宽30

```

```
- funding列加`$`货币格式
- avg_salary列加`$`保留两位小数
...
```

专业Excel导出完成!

Out[53]: '\n**代码做了什么:**\n- 深蓝色表头 + 白色加粗字体\n- 冻结首行, 滚动时表头不动\n- 列宽自适应, 最宽30\n- funding列加`\$`货币格式\n- avg_salary列加`\$`保留两位小数\n'

最后来一步骤, 简单做一个线性回归预测模型! 目的在于预测所有的选区, 记得不是机构! 总共190多个选区下, 根据17-25年所有数据, 来分析每个选区的未来funding走势来判断后续可以理论性重点考虑投资哪一个选区最为合适?

In [54]: stg_cs_j

Out[54]:

	year	region	constituency	org_name	amount	jobs
0	2025	Alberta	Banff - Airdrie	Airdrie & District Soccer Association	31621	7
1	2025	Alberta	Banff - Airdrie	Airdrie Chamber of Commerce	9341	2
2	2025	Alberta	Banff - Airdrie	Alberta Institute for Wildlife Conservation	74421	16
3	2025	Alberta	Banff - Airdrie	Ascend Pediatrics	1800	1
4	2025	Alberta	Banff - Airdrie	Banff Sport Medicine Foundation	4708	1
...	...	...	...	...	...	...
137516	2017	Alberta	Yellowhead	Yellow Submarine	6043	3
137517	2017	Alberta	Yellowhead	Yellowhead County Library Board	7394	2
137518	2017	Alberta	Yellowhead	Yellowhead Gas Co-Op Ltd	6510	1
137519	2017	Alberta	Yellowhead	fRI Research	4392	1
137520	2017	Alberta	Summer Work Experience	McMurray Métis	290210	50

137521 rows × 6 columns

In [55]: # 按照省份, 选区, 年来分组!
constituency_yearly = stg_cs_j.groupby(['region', 'constituency', 'year']).agg(tota

In [56]: constituency_yearly

Out[56]:

	region	constituency	year	total_funding	total_jobs
0	Alberta	Banff - Airdrie	2017	369399	117
1	Alberta	Banff - Airdrie	2018	371332	132
2	Alberta	Banff - Airdrie	2019	731178	215
3	Alberta	Banff - Airdrie	2020	1038083	234
4	Alberta	Banff - Airdrie	2021	1590778	467
...	...	...	...	...	...
1769	Ontario	York South - Weston	2021	1193418	265
1770	Ontario	York South - Weston	2022	1229586	270
1771	Ontario	York South - Weston	2023	899360	201
1772	Ontario	York South - Weston	2024	880893	181
1773	Ontario	York South - Weston	2025	1388101	282

1774 rows × 5 columns

In [57]: constituency_yearly.head(50)

Out[57]:

	region	constituency	year	total_funding	total_jobs
0	Alberta	Banff - Airdrie	2017	369399	117
1	Alberta	Banff - Airdrie	2018	371332	132
2	Alberta	Banff - Airdrie	2019	731178	215
3	Alberta	Banff - Airdrie	2020	1038083	234
4	Alberta	Banff - Airdrie	2021	1590778	467
5	Alberta	Banff - Airdrie	2022	1281065	375
6	Alberta	Banff - Airdrie	2023	748275	211
7	Alberta	Banff - Airdrie	2024	913255	271
8	Alberta	Banff - Airdrie	2025	613802	201
9	Alberta	Battle River - Crowfoot	2017	860600	263
10	Alberta	Battle River - Crowfoot	2018	620996	213
11	Alberta	Battle River - Crowfoot	2019	807424	228
12	Alberta	Battle River - Crowfoot	2020	1365890	336
13	Alberta	Battle River - Crowfoot	2021	2086772	606
14	Alberta	Battle River - Crowfoot	2022	1284902	383
15	Alberta	Battle River - Crowfoot	2023	760788	236
16	Alberta	Battle River - Crowfoot	2024	840178	252
17	Alberta	Battle River - Crowfoot	2025	906024	275
18	Alberta	Bow River	2017	475771	142
19	Alberta	Bow River	2018	367059	132
20	Alberta	Bow River	2019	644654	187
21	Alberta	Bow River	2020	645109	154
22	Alberta	Bow River	2021	1346486	390
23	Alberta	Bow River	2022	1155966	369
24	Alberta	Bow River	2023	634233	174
25	Alberta	Bow River	2024	682787	195
26	Alberta	Bow River	2025	644145	186
27	Alberta	Calgary Centre	2017	1026516	270
28	Alberta	Calgary Centre	2018	805222	228
29	Alberta	Calgary Centre	2019	1343501	292
30	Alberta	Calgary Centre	2020	1603933	404
31	Alberta	Calgary Centre	2021	3060723	867
32	Alberta	Calgary Centre	2022	1820100	506
33	Alberta	Calgary Centre	2023	1131693	304

	region	constituency	year	total_funding	total_jobs
34	Alberta	Calgary Centre	2024	1155088	235
35	Alberta	Calgary Centre	2025	1204981	334
36	Alberta	Calgary Confederation	2017	966045	301
37	Alberta	Calgary Confederation	2018	987397	338
38	Alberta	Calgary Confederation	2019	1342816	420
39	Alberta	Calgary Confederation	2020	1715197	423
40	Alberta	Calgary Confederation	2021	3238924	944
41	Alberta	Calgary Confederation	2022	2839088	820
42	Alberta	Calgary Confederation	2023	1429442	397
43	Alberta	Calgary Confederation	2024	1467792	417
44	Alberta	Calgary Confederation	2025	1591402	500
45	Alberta	Calgary Forest Lawn	2017	382729	122
46	Alberta	Calgary Forest Lawn	2018	438357	150
47	Alberta	Calgary Forest Lawn	2019	555432	154
48	Alberta	Calgary Forest Lawn	2020	633048	130
49	Alberta	Calgary Forest Lawn	2021	1143113	281

```
In [58]: from sklearn.linear_model import LinearRegression
import numpy as np

results = []

#每个选区group单独训练模型!
for (region, constituency), group in constituency_yearly.groupby(['region', 'constituency']):
    X = group['year'].values.reshape(-1,1) # year as features!
    y_funding = group['total_funding'].values # funding as the target 1
    y_jobs = group['total_jobs'].values # jobs created as the target 2

    # train the model
    model_funding = LinearRegression().fit(X, y_funding)
    model_jobs = LinearRegression().fit(X, y_jobs)

    # predict for values in 2026
    pred_funding = model_funding.predict([[2026]])[0]
    pred_jobs = model_jobs.predict([[2026]])[0]

    results.append({
        'region': region,
        'constituency': constituency,
        'pred_funding_2026': round(max(pred_funding, 0)),
        'pred_jobs_2026': round(max(pred_jobs, 0))
    })

predictions = pd.DataFrame(results)
print(predictions.head(10))
```

	region	constituency	pred_funding_2026	pred_jobs_2026
0	Alberta	Banff - Airdrie	1090843	321
1	Alberta	Battle River - Crowfoot	1114701	329
2	Alberta	Bow River	908803	260
3	Alberta	Calgary Centre	1590974	416
4	Alberta	Calgary Confederation	2167547	622
5	Alberta	Calgary Forest Lawn	1235810	333
6	Alberta	Calgary Heritage	1342974	321
7	Alberta	Calgary Midnapore	1446217	404
8	Alberta	Calgary Nose Hill	940869	223
9	Alberta	Calgary Rocky Ridge	1266578	342

In [60]: `predictions.shape`

Out[60]: (198, 4)

```
In [61]: with pd.ExcelWriter(output_path, mode='a', engine='openpyxl') as writer:
          predictions.to_excel(writer, sheet_name='2026 Predictions', index=False)

          print("预测sheet添加完成! ")
```

预测sheet添加完成!

In [62]: `all_ridings_long`

Out[62]:

	metric	constituency	province	total_number_of_organizations	year	avg_salary	
	0	Abbotsford	British Columbia	208	2017	3653.07	
	1	Abbotsford	British Columbia	208	2018	2690.23	
	2	Abbotsford	British Columbia	208	2019	4243.39	
	3	Abbotsford	British Columbia	208	2020	4154.58	
	4	Abbotsford	British Columbia	208	2021	3773.36	
	...	...	...	...	...	...	
	1777	York South - Weston	Ontario	243	2021	4503.46	1
	1778	York South - Weston	Ontario	243	2022	4554.02	1
	1779	York South - Weston	Ontario	243	2023	4474.43	
	1780	York South - Weston	Ontario	243	2024	4866.81	
	1781	York South - Weston	Ontario	243	2025	4922.34	1

1782 rows × 7 columns



```
In [64]: # 把predictions整理成和all_ridings_long一样的行格式
pred_rows = predictions.copy()
pred_rows = pred_rows.rename(columns={
    'region': 'province',
    'pred_funding_2026': 'funding',
    'pred_jobs_2026': 'jobs'
})
pred_rows['year'] = 2026
pred_rows['avg_salary'] = (pred_rows['funding'] / pred_rows['jobs']).round(2)
pred_rows['total_number_of_organizations'] = None
pred_rows['is_predicted'] = True

# 原始数据加标记
all_ridings_long['is_predicted'] = False

# 拼接
all_ridings_long = pd.concat(
    [all_ridings_long, pred_rows[all_ridings_long.columns]],
    ignore_index=True
)

print(all_ridings_long.shape) # 应该是1980行

(1980, 8)
```

```
In [65]: all_ridings_long
```

Out[65]:

	constituency	province	total_number_of_organizations	year	avg_salary	1
0	Abbotsford	British Columbia	208	2017	3653.07	5
1	Abbotsford	British Columbia	208	2018	2690.23	3
2	Abbotsford	British Columbia	208	2019	4243.39	5
3	Abbotsford	British Columbia	208	2020	4154.58	5
4	Abbotsford	British Columbia	208	2021	3773.36	7
...	...	...	...	...	...	...
1975	Windsor - Tecumseh	Ontario	None	2026	4231.10	14
1976	Windsor West	Ontario	None	2026	4772.74	17
1977	York - Simcoe	Ontario	None	2026	4719.38	11
1978	York Centre	Ontario	None	2026	5326.29	10
1979	York South - Weston	Ontario	None	2026	5144.05	13

1980 rows × 8 columns



```
In [66]: all_ridings_long.to_csv(
    r'C:\Users\admin\Desktop\YRES\Andi Dong - Summer Job Analysis\tableau_riding
    index=False
)
print("导出完成! ")
```

导出完成!

Python Layer - CSJ Analysis Pipeline

目标: SQL做不到的自动化处理, 为Tableau和YRES交付做准备

Step 1: 连接DuckDB, 读取6张mart表

- 连接 `csj_analysis.duckdb`
- 读取 `mart_on/bc/ab_ridings` 和 `mart_on/bc/ab_organizations`
- 验证数据形状: ON(121,29) BC(42,29) AB(35,29)

Step 2: 数据清洗

- 找出所有 `avg_salary` 列
 - 统一四舍五入到2位小数
 - 用for loop批量处理6张表
-

Step 3: 加province列 + concat拼接

- 6张表各自加 `province` 列标记省份
 - riding三表合并 → `all_ridings` (198行 × 30列)
 - org三表合并 → `all_orgs` (5911行 × 30列)
-

Step 4: 宽表转长表 (pd.melt + pivot_table)

为什么要转：Tableau需要year作为独立列才能画趋势图

- `pd.melt()` 把年份列转成行
 - `str.rsplit('_', n=1)` 拆分metric和year
 - `pivot_table()` 把funding/jobs/avg_salary变回三列
 - 结果：
 - `all_ridings_long` (1782行 × 7列)
 - `all_orgs_long` (53199行 × 7列)
-

Step 5: 线性回归预测2026

为什么用线性回归：每个选区只有9年数据，复杂模型会过拟合

- 从 `stg_csj` 用pandas groupby得到选区年度汇总
 - 对198个选区各自训练独立的LinearRegression模型
 - $X = \text{year}, y = \text{total_funding} / \text{total_jobs}$
 - 预测2026年每个选区的funding和jobs
 - `max(pred, 0)` 确保预测值不为负
-

Step 6: 预测结果整合

- 预测结果加入Excel作为第7个sheet (`2026 Predictions`)
 - 预测结果转成长表格式，加 `is_predicted=True` 标记
 - 拼接进 `all_ridings_long`
 - 最终：1980行 (1782历史 + 198预测)
-

Step 7: 导出Excel (YRES交付)

- 6个sheet对应6张分析表

- 格式：深蓝表头 + 白色加粗字体 + 冻结首行 + 列宽自适应
- funding列： `$#,##0` 货币格式
- avg_salary列： `$#,##0.00` 格式
- 第7个sheet：2026预测结果

Step 8: 导出Tableau CSV (3个文件)

文件名	内容	用途
tableau_ridings_long.csv	198选区 × 10年(含2026预测)	趋势图/预测对比
tableau_orgs_top30_long.csv	TOP30组织历史数据	竞争对手分析
tableau_all_orgs_full.csv	全量原始数据	深度分析

技术栈总结

- duckdb

pandas

openpyxl

sklearn
- 连接数据库

→ 数据处理 (groupby/melt/pivot/concat/merge)

→ Excel格式美化

→ 线性回归预测

In []: